Bansilal Ramnath Agarwal Charitable Trust's

VISHWAKARMA INSTITUTE OF TECHNOLOGY – PUNE

Software Engineering

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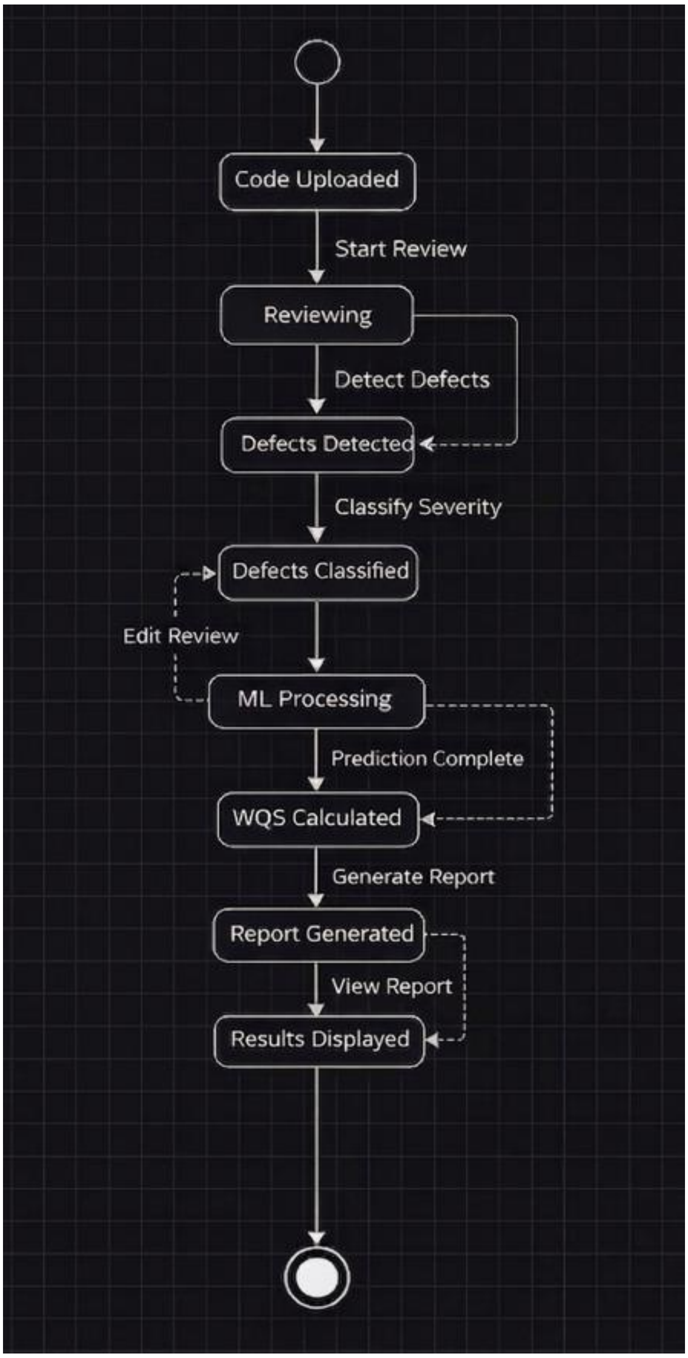
Assignment-6

Model the dynamic behavior of a system using state diagrams for Course Project.



1. **Conceptual Overview**

The provided State Machine Diagram illustrates the dynamic behavior of a Code Review System. In UML (Unified Modeling Language), a state diagram focuses on the lifecycle of a single object—in this case, a Code Submission—and how it changes from one "state" to another based on specific triggers or events.

2. **State Transition Analysis**

The system logic is defined by the following components:

- Initial & Final States: The process begins at the Initial Pseudo-state (top circle) when a user initiates an upload and terminates at the Final State (bottom bullseye) once results are viewed.
- Sequential Progression: The system follows a logical progression from raw data ingest (Code Uploaded) to analytical processing.
- Conditional Loops: * The Review Loop: The transition between *Reviewing* and *Defects Detected* represents an iterative discovery phase where the system identifies issues.
- The Feedback Loop: The "Edit Review" transition between *ML Processing* and *Defects Classified* represents a "Human-in-the-Loop" mechanism, allowing for manual correction or refinement before final calculations are performed.
- Data Synthesis: The WQS Calculated (Weighted Quality Score) state is a critical juncture where the system aggregates manual classifications and machine learning predictions into a standardized metric.

3. Design Justification

Modeling the system using a state diagram ensures that the software architecture handles transitions safely. By defining states like ML Processing and WQS Calculated, the system ensures that no report is generated until the computational logic is complete, thereby maintaining data integrity and process synchronization.